

Part A: Theory & Definitions

1. Define the following with examples from the dataset:

Types of data: Numerical & Categorical

Data in statistics is mainly divided into **Numerical** and **Categorical** data.

◆ Numerical Data

Numerical data consists of values that can be **measured or counted** and are expressed in numbers. Mathematical operations (addition, average, etc.) can be performed on them.

Examples from the dataset:

- **Age_of_Household_Head** → e.g., 35, 52, 68
- **Household_Income** → e.g., 25,000; 60,000; 1,10,000
- **Family_Size** → e.g., 2, 4, 6

✓ These values represent quantities and can be used to calculate mean, median, variance, etc.

◆ Categorical Data

Categorical data represents **labels, groups, or categories**. It describes qualities or characteristics and cannot be used for mathematical calculations.

Examples from the dataset:

- **Household_ID** → HH0001, HH0025
- **Education_Level** → Primary, Secondary, Graduate, Post-Grad
- **Owns_House** → Yes / No
- **Urban_Rural** → Urban / Rural

✓ These values classify households into different groups.

Types of Statistics: Descriptive vs. Inferential

Statistics is broadly classified into **Descriptive Statistics** and **Inferential Statistics**.

◆ Descriptive Statistics

Descriptive statistics focuses on **summarizing and organizing data** to understand its main features.

Examples from the dataset:

- Average household income
- Maximum and minimum age of household heads
- Count of urban vs rural households
- Percentage of households owning a house

✓ It describes *what the data shows*.

◆ Inferential Statistics

Inferential statistics uses **sample data** to make **predictions or conclusions about a population**.

Examples related to the dataset:

- Predicting average income of all households based on the 500-sample data
- Testing whether urban households earn more than rural households
- Estimating population income using confidence intervals

✓ It explains *what we can conclude beyond the data*.

What is Descriptive Statistics?

Definition:

Descriptive statistics is the branch of statistics that deals with **collecting, organizing, summarizing, and presenting data** in a meaningful way.

Common descriptive measures:

- Mean, Median, Mode
- Minimum, Maximum
- Standard Deviation
- Frequency tables and graphs

Example from the dataset:

- Calculating the **average age** of household heads
- Finding the **most common education level**
- Displaying **income distribution using a histogram**
- Counting how many households own a house

✓ Descriptive statistics helps us **understand the dataset without making predictions.**

2. Explain the difference between:

Mean, Median, Mode

Range, Variance, Standard Deviation

A) Mean, Median, and Mode

These are **measures of central tendency** (they show the center of the data).

◆ Mean

Definition:

The **mean** is the average value of a dataset.

Formula:

Mean = (Sum of all values) ÷ (Number of values)

Example (Household_Income):

If incomes are: 20,000; 30,000; 50,000

Mean = (20,000 + 30,000 + 50,000) / 3 = **33,333**

✓ Sensitive to very high or very low values.

◆ Median

Definition:

The **median** is the **middle value** when data is arranged in ascending or descending order.

Example (Age_of_Household_Head):

Ages: 30, 35, **40**, 45, 60

Median = **40**

✓ Not affected by extreme values (outliers).

◆ Mode

Definition:

The **mode** is the value that appears **most frequently** in the dataset.

Example (Education_Level):

If “Secondary” appears most often → **Mode = Secondary**

✓ Can be used for categorical data.

✦ Key Differences (Mean vs Median vs Mode)

Measure	Meaning	Affected by Outliers	Example Field
Mean	Average	Yes	Household_Income
Median	Middle value	No	Age_of_Household_Head
Mode	Most frequent	No	Education_Level

B) Range, Variance, and Standard Deviation

These are **measures of dispersion** (they show data spread).

◆ Range

Definition:

Range is the difference between the **maximum and minimum** values.

Formula:

Range = Max – Min

Example (Family_Size):

Max = 7, Min = 1

Range = 6

✓ Very simple but ignores overall distribution.

◆ Variance

Definition:

Variance measures how far each value is from the **mean**, on average.

Formula (Conceptual):

Variance = Average of squared deviations from the mean

Example (Income):

Large variance → incomes are widely spread

Small variance → incomes are similar

✓ Units are **squared**, making interpretation harder.

◆ **Standard Deviation**

Definition:

Standard deviation is the **square root of variance**.

Why important?

It shows data spread in the **same unit as the data**.

Example (Household_Income):

SD = 10,000 → most incomes vary by about ₹10,000 from the mean

✓ Most commonly used dispersion measure.

✦ **Key Differences (Range vs Variance vs Standard Deviation)**

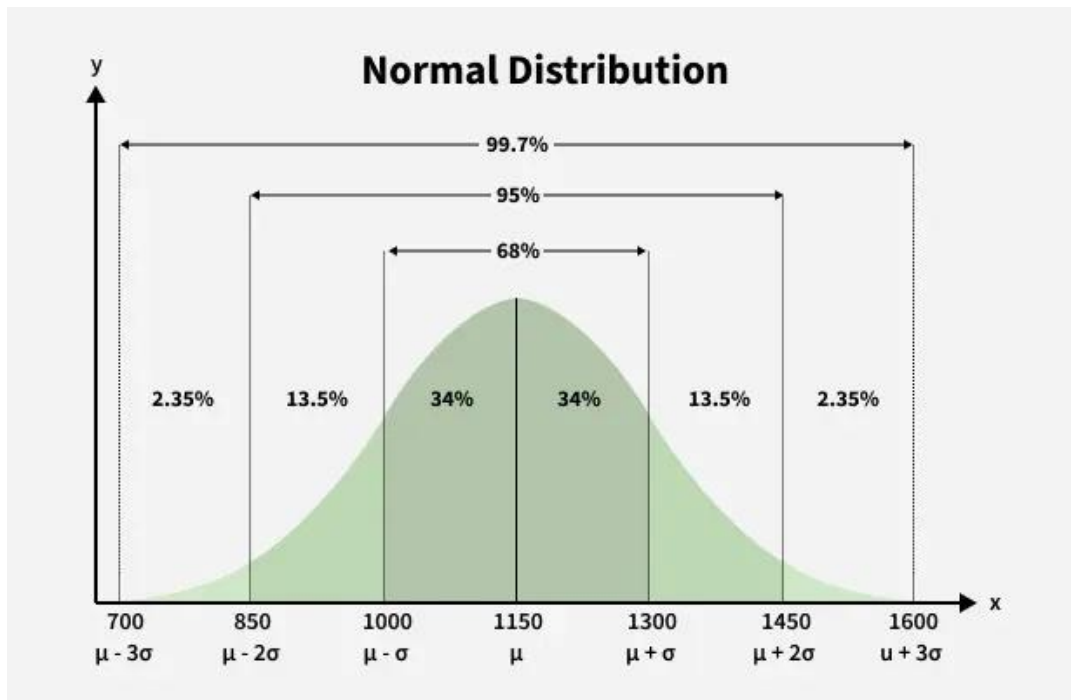
Measure	What it Measures	Unit
Range	Total spread	Same as data
Variance	Average squared spread	Squared unit
Standard Deviation	Average spread from mean	Same as data

3. Explain the following term with neat and clean diagram along with its formula:

Gaussian Distribution**Definition**

A **Gaussian Distribution**, also called a **Normal Distribution**, is a **continuous probability distribution** that is **symmetrical and bell-shaped** around its mean.

Most natural and real-life data (such as **height, marks, IQ, measurement errors**) approximately follow a Gaussian distribution.



Key Characteristics

- Bell-shaped and **symmetric** curve
- **Mean = Median = Mode**
- The curve is **highest at the mean**
- Total area under the curve = **1**
- Tails approach but never touch the X-axis

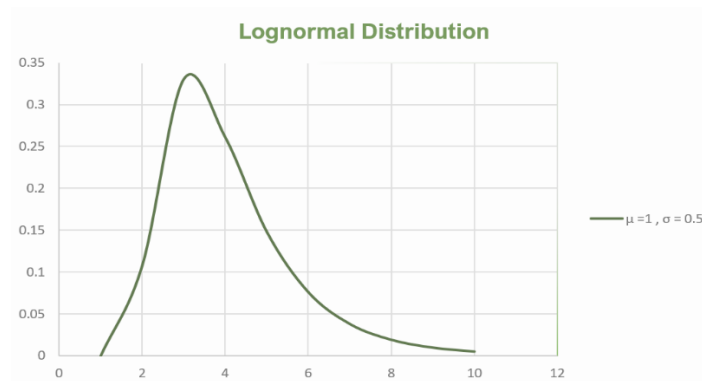
Gaussian Distribution Formula

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Where:

Symbol	Meaning
x	Random variable
μ	Mean
σ	Standard Deviation
π	Constant (≈ 3.1416)
e	Euler's number (≈ 2.718)

Log Normal Distribution



Definition

A **Log-Normal Distribution** is a **continuous probability distribution** in which the **logarithm of the random variable follows a normal (Gaussian) distribution**.

If

$$Y = \ln (X)$$

is **normally distributed**, then **X is log-normally distributed**.

Key Characteristics

- **Positively skewed (right-skewed)**
- Values are **always positive** ($X > 0$)
- Mean > Median > Mode
- Common in **real-world growth processes**

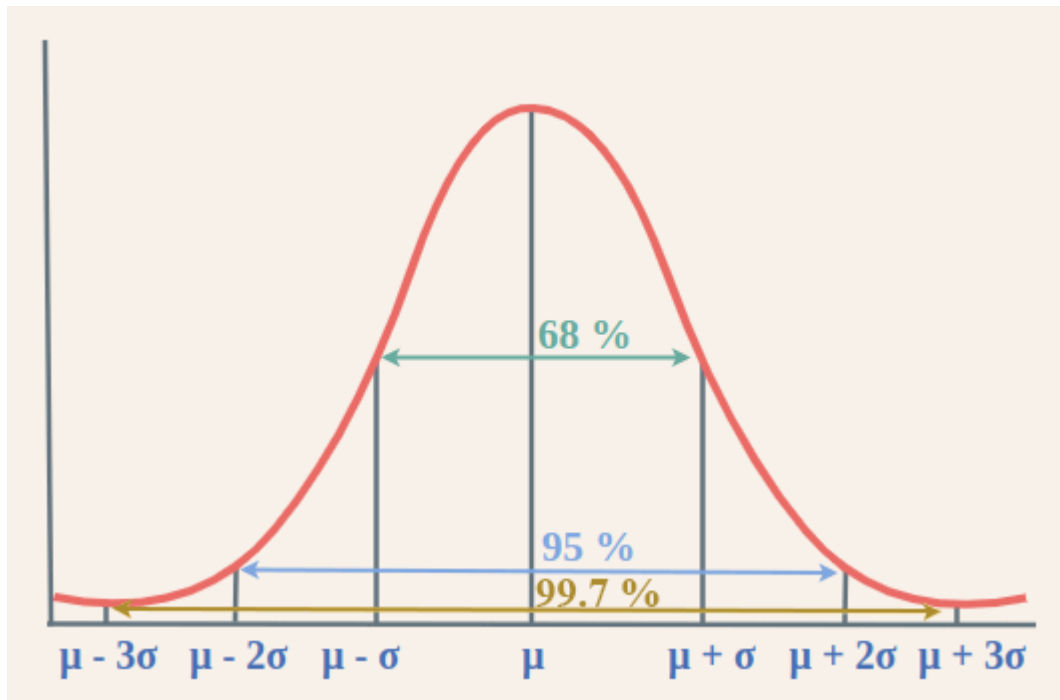
Log-Normal Distribution Formula

$$f(x) = \frac{1}{x \sigma \sqrt{2\pi}} \exp \left(-\frac{(\ln x - \mu)^2}{2\sigma^2} \right), x > 0$$

Where:

Symbol	Meaning
x	Random variable
μ	Mean of $\ln (x)$
σ	Standard deviation of $\ln (x)$
$\ln$	Natural logarithm
e	Euler's constant

3-Sigma Rule or Empirical Rule



Definition

The **3-Sigma Rule**, also known as the **Empirical Rule**, states that for a **normally distributed (Gaussian) dataset**, **almost all observations lie within three standard deviations of the mean**.

Statement of the Rule

For a **Normal Distribution**:

Range Percentage of Data

$$\mu \pm 1\sigma \approx 68\%$$

$$\mu \pm 2\sigma \approx 95\%$$

$$\mu \pm 3\sigma \approx 99.7\%$$

This is why it is called the **68–95–99.7 Rule**.

Percentiles

Definition

A **percentile** is a measure that indicates the **value below which a given percentage of observations in a dataset falls**.

👉 The p^{th} percentile is the value such that **p% of the data is less than or equal to it.**

Key Points

- Used to understand **relative position** in a dataset
 - Divides data into **100 equal parts**
 - Helps compare individual performance
-

Percentile Formula (Ungrouped Data)

$$\text{Position of } P_p = \frac{p}{100} \times (n + 1)$$

Where:

- p = required percentile
 - n = total number of observations
-

Steps to Find Percentile

1. Arrange data in **ascending order**
 2. Calculate position using formula
 3. If position is fractional, **interpolate** between values
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Example

Data (sorted):

10, 20, 30, 40, 50, 60, 70, 80, 90

Find **40th percentile**:

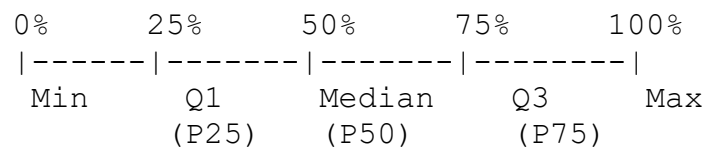
$$P_{40} = \frac{40}{100} \times (9 + 1) = 4$$

👉 4th value = **40**

So, **40 is the 40th percentile.**

Neat Diagram

Data Distribution (Ordered)



Important Percentiles

Percentile	Meaning
P25	First Quartile (Q1)
P50	Median
P75	Third Quartile (Q3)
P90	Top 10% cutoff
P95	High performers

Quartiles

Definition

Quartiles divide an **ordered dataset** into **four equal parts**, with each part containing **25% of the data**.

They help in understanding **data spread, central tendency, and variability**.

Types of Quartiles

Quartile	Symbol	Meaning
First Quartile	Q_1	25% of data lies below
Second Quartile	Q_2	50% of data lies below (Median)
Third Quartile	Q_3	75% of data lies below

Quartile Formula (Ungrouped Data)

$$Q_1 = \frac{(n+1)}{4}$$
$$Q_2 = \frac{(n+1)}{2}$$
$$Q_3 = \frac{3(n+1)}{4}$$

Where:

- n = number of observations
-

Steps to Find Quartiles

1. Arrange data in **ascending order**
 2. Calculate positions of Q_1 , Q_2 , Q_3
 3. If position is fractional, **interpolate**
-

Example

Data (sorted):

10, 20, 30, 40, 50, 60, 70, 80, 90

- $n = 9$

$$Q_1 = \frac{10}{4} = 2.5 \Rightarrow \frac{20 + 30}{2} = 25$$

$$Q_2 = \frac{10}{2} = 5 \Rightarrow 50$$

$$Q_3 = \frac{30}{4} = 7.5 \Rightarrow \frac{70 + 80}{2} = 75$$

Neat Diagram

Ordered Data Distribution

Min	Q1	Q2	Q3	Max
----- ----- ----- -----				
	25%	50%	75%	

Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

👉 Measures **spread of middle 50% data**

Outlier Detection Using Quartiles

Lower Bound = $Q_1 - 1.5 \times IQR$

Upper Bound = $Q_3 + 1.5 \times IQR$

Values outside this range are **outliers**.

Five Number Summary

Definition

The **Five Number Summary** is a set of **five descriptive statistics** that provide a **complete overview of a dataset’s distribution**, including its **center, spread, and range**.

The Five Numbers

Statistic	Symbol	Meaning
Minimum	Min	Smallest value
First Quartile	Q_1	25% of data lies below
Median	Q_2	Middle value (50%)
Third Quartile	Q_3	75% of data lies below
Maximum	Max	Largest value

Steps to Find Five Number Summary

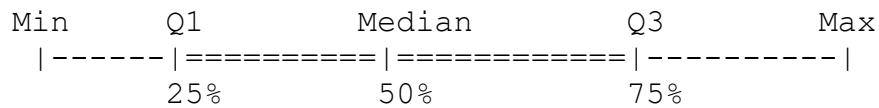
1. Arrange data in **ascending order**
2. Identify **Minimum** and **Maximum**
3. Find **Q_1** , **Median (Q_2)**, and **Q_3** using quartile formulas

Example

Data (sorted):
10, 20, 30, 40, 50, 60, 70, 80, 90

- **Min** = 10
- **Q_1** = 25
- **Median (Q_2)** = 50
- **Q_3** = 75
- **Max** = 90

Neat Diagram (Box-Plot Representation)



Interquartile Range (IQR)

$$IQR = Q_3 - Q_1$$

👉 Measures spread of the **middle 50% of data**

Outlier Detection

Using IQR method:

$$\begin{aligned}\text{Lower Bound} &= Q_1 - 1.5 \times IQR \\ \text{Upper Bound} &= Q_3 + 1.5 \times IQR\end{aligned}$$

Values outside this range are **outliers**.

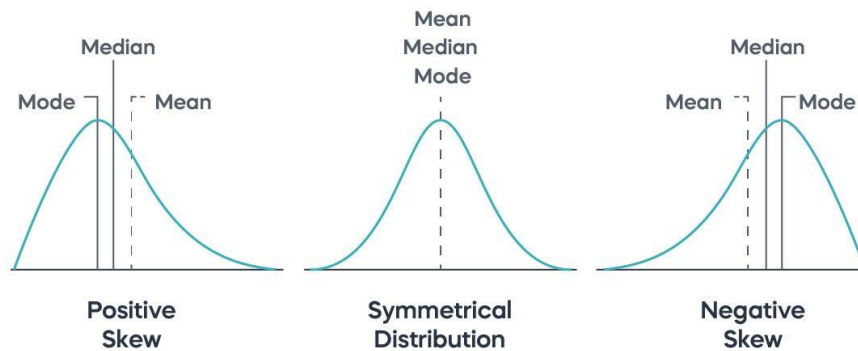
Why Five Number Summary is Important

- Gives **quick overview** of data
 - Helps detect **skewness & outliers**
 - Basis for **box plots**
 - Easy comparison between datasets
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Applications

- Exploratory Data Analysis (EDA)
- Exam score analysis
- Salary distribution
- Data science & statistics

Skewness



Definition

Skewness is a **measure of asymmetry** of a probability distribution around its mean.

It tells us **whether the data is symmetric, right-skewed, or left-skewed**.

Types of Skewness

Type	Description
Zero Skewness	Symmetric distribution
Positive Skewness	Right-skewed (long right tail)
Negative Skewness	Left-skewed (long left tail)

Skewness Formula

Karl Pearson's Coefficient

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

If mode is not available:

$$\text{Skewness} = \frac{3(\text{Mean} - \text{Median})}{\text{Standard Deviation}}$$

Moment Coefficient of Skewness

$$\gamma_1 = \frac{\frac{1}{n} \sum (x_i - \bar{x})^3}{\sigma^3}$$

Interpretation

Value	Meaning
Sk = 0	Symmetric
Sk > 0	Positively skewed
Sk < 0	Negatively skewed

Example

- **Income distribution** → Positive skewness
 - **Exam scores (very easy paper)** → Negative skewness
 - **Height of students** → Approximately symmetric
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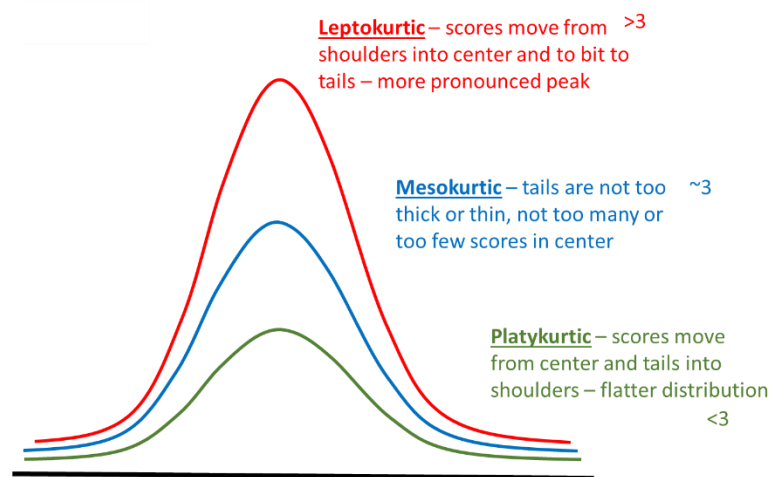
Why Skewness is Important

- Helps understand **data shape**
 - Influences **mean & median**
 - Important in **statistical modeling**
 - Helps choose correct **statistical tests**
-

Relation with Mean, Median, Mode

Distribution	Relation
Symmetric	Mean = Median = Mode
Positive skew	Mean > Median > Mode
Negative skew	Mean < Median < Mode

Kurtosis



Definition

Kurtosis is a **statistical measure that describes the peakedness and tail heaviness** of a probability distribution **relative to a normal distribution**.

👉 It tells us **how extreme values (outliers) are likely to occur**.

Types of Kurtosis

Type	Description
Mesokurtic	Normal peak (Gaussian)
Leptokurtic	Sharp peak, heavy tails
Platykurtic	Flat peak, light tails

Skewness Formula

Karl Pearson's Coefficient

$$\text{Skewness} = \frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$$

If mode is not available:

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Moment Coefficient of Skewness

$$\gamma_1 = \frac{\frac{1}{n} \sum (x_i - \bar{x})^3}{\sigma^3}$$

Interpretation

Value Meaning

Sk = 0 Symmetric

Sk > 0 Positively skewed

Sk < 0 Negatively skewed

Example

- **Income distribution** → Positive skewness
 - **Exam scores (very easy paper)** → Negative skewness
 - **Height of students** → Approximately symmetric
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Why Skewness is Important

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Relation with Mean, Median, Mode

Distribution	Relation
Symmetric	Mean = Median = Mode
Positive skew	Mean > Median > Mode
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